

MAML-Select: An Online Adaptive Client Selection Method for Federated Learning via Meta-Learning

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We sincerely thank the Editor, Associate Editor, and the anonymous Reviewers for their valuable time and effort in evaluating our manuscript. We have carefully addressed all comments and suggestions in the revised version. The manuscript has been revised for clarity, technical completeness, experimental evidence, and presentation quality. [All textual additions and revisions made in response to the Reviewers' comments are highlighted in blue in the marked manuscript.](#) A separate clean black-and-white manuscript is also provided for normal reading. Please note that the Reviewers' comments are *italicized*, followed by our replies in normal font.

Summary of Major Revisions

- We clarified the novelty: MAML-Select meta-learns and rapidly adapts the client-selection policy, rather than personalizing client models or learning a long-horizon reinforcement-learning controller.
- We added a direct comparison table for prior work and baselines.
- We expanded the evaluation to Fashion-MNIST, CIFAR-10, and CIFAR-100 under non-IID Dirichlet $\alpha = 0.5$ with $N = 100$, $K = 10$, $E = 5$, batch size 32, and the seed protocol reported in the revised manuscript.
- We added FedGCS and CriticalFL to the revised comparison framework.
- We now report mean $\pm$ standard deviation, paired tests, λ sensitivity, state-feature ablation, fairness and coverage, modelled energy and carbon, and selector-overhead scaling.
- We clarified the resource-accounting scope in the main manuscript. The revised paper states the device-tier power values, battery capacities, and grid intensity used to compute modelled energy/carbon.

Response to Comments of Reviewer 1

Comment 1: *Simplify long and complex sentences, especially in the Introduction and Related Work.*

Reply 1: We agree with the Reviewer. We revised the full manuscript for shorter sentence structure and clearer narrative flow. The Abstract, Impact Statement, Introduction, Related Work, Methodology, Evaluation, and Conclusion were all edited.

Changes incorporated in the revised manuscript: Long sentences were divided and the central motivation is now stated directly: client utility changes during FL training, so a fixed selection rule can become stale.

Comment 2: *Provide more implementation details, including optimizer type, learning rates, schedules, exact meta-policy architecture, initialization, random seeds, and code or detailed pseudocode.*

Reply 2: We thank the Reviewer for this important suggestion. We added the requested implementation details. The policy is a 6-64-64-1 MLP with ReLU activations and 4,673 trainable parameters. The local models use PyTorch default initialization and SGD with learning rate 0.01, momentum 0.9, weight decay 10^{-4} , and a constant scheduler. The MAML policy uses one inner step with $\beta = 0.01$ and Adam for the outer update with $\eta = 0.001$. Seeds are 42, 123, and 2026, and the selector seed is fixed to 2026.

Changes incorporated in the revised manuscript: The Methodology now includes Algorithm 1 with explicit support adaptation, candidate scoring, top- K selection, local-loss feedback routing, and first-order meta-update. The Evaluation section now reports optimizer, learning-rate, scheduler, initialization, seed, and meta-policy settings.

Comment 3: *The accuracy gains are modest. Please justify why they are meaningful and identify scenarios where efficiency is prioritized.*

Reply 3: We appreciate this comment and have reframed the contribution as an efficiency–accuracy trade-off, not as universal accuracy dominance. Across shared FedAvg seeds, MAML-Select changes accuracy by -0.11 , -3.81 , and -1.51 percentage points on Fashion-MNIST, CIFAR-10, and CIFAR-100, respectively, while reducing TFLOPs by 12.8%, 21.5%, and 1.7%, and modelled energy/carbon by 16.4%, 24.7%, and 4.7%. This is meaningful in edge settings with device limits, intermittent connectivity, and strict round deadlines.

Changes incorporated in the revised manuscript: The Abstract, Introduction, Evaluation, and Conclusion now describe MAML-Select as a resource-efficient selector with competitive accuracy and explicitly note the accuracy trade-off on CIFAR-10.

Response to Comments of Reviewer 2

Comment 1: *Clarify the exact novelty compared with RL-based approaches, contextual bandits, and meta-learning-based personalization frameworks. Add a dedicated comparison table.*

Reply 1: We thank the Reviewer for this suggestion. We clarified the distinction. RL and bandit approaches

learn or estimate selection rewards through interaction. Personalization methods use meta-learning to adapt model parameters for each client. MAML-Select instead meta-learns the initialization of the client-ranking policy itself and performs a gradient-based adaptation step from recent FL feedback before each selection decision.

Changes incorporated in the revised manuscript: The Introduction, Related Work, and Table I now compare FedAvg, FedCS, Oort, TiFL, FedCor, CriticalFL, FedGCS, personalization-based meta-learning, and MAML-Select across objective, adaptation mechanism, dominant operation, and optimization strategy.

Comment 2: *The manuscript has empirical evaluation only. There is no convergence proof or theoretical justification for non-stationary adaptation.*

Reply 2: We agree that a theoretical justification is useful. We added a limited theoretical statement. It is intentionally local: under an L -smooth support loss and a standard step-size condition, the inner update decreases the support objective. This supports the selector-level adaptation step used by MAML-Select. We do not claim a full global FL convergence theorem.

Changes incorporated in the revised manuscript: A descent statement and interpretation now appear immediately after the inner-loop equation. The text explicitly labels it as a selector-level descent property, not a complete FL convergence proof.

Comment 3: *Explain the manually chosen trade-off parameter λ and evaluate sensitivity.*

Reply 3: We added both explanation and evidence. The manuscript states that λ controls the latency–utility trade-off. Larger λ emphasizes latency and resource efficiency. The new sensitivity analysis sweeps $\lambda \in \{0.1, 0.5, 1.0, 5.0\}$ on Fashion-MNIST. Accuracy remains stable across the tested range, while compute and energy decrease as the resource term receives more weight. The fairness curve is more sensitive at larger λ , which explains why $\lambda = 0.5$ is used as a balanced default rather than an extreme low-cost setting.

Changes incorporated in the revised manuscript: The System Model explains λ . The manuscript now reports the λ sweep next to the corresponding sensitivity text and states why $\lambda = 0.5$ is used as the default operating point.

Comment 4: *The experimental validation is limited to small-scale datasets. Add larger or more realistic FL benchmarks.*

Reply 4: We expanded the evaluation from two datasets to three by adding CIFAR-100 with ResNet18. CIFAR-100 provides a harder 100-class non-IID benchmark than Fashion-MNIST and CIFAR-10. Fashion-MNIST and CIFAR-10 are reported for 200 rounds, while CIFAR-100 is analyzed for 150 rounds because it is the most computationally demanding setting. We also added a selector-overhead scaling probe with $N \in \{20, 40, 80, 100\}$ and $K/N = 0.1$.

Changes incorporated in the revised manuscript: The Evaluation now includes Fashion-MNIST, CIFAR-10, and CIFAR-100. The main manuscript also includes the selector time-complexity proof and a selector-

overhead scaling plot.

Comment 5: *Include standard deviation across multiple runs and statistical significance analysis.*

Reply 5: We agree. The main benchmark table now reports mean $\pm$ standard deviation for the repeated-run comparisons. We also computed paired tests over shared seeds. Against FedAvg, MAML-Select reduces TFLOPs, modelled energy, and carbon on all three datasets. The paired tests give $p < 0.05$ for these resource reductions. The TFLOP p -values are 0.009, 0.002, and 0.012, and the energy/carbon p -values are 0.007, 0.001, and 0.003 for Fashion-MNIST, CIFAR-10, and CIFAR-100, respectively. Accuracy is reported as the corresponding efficiency–accuracy trade-off rather than as universal accuracy dominance.

Changes incorporated in the revised manuscript: Table II now reports accuracy, weighted precision, recall, F1, TFLOPs, modelled energy/carbon, Jain fairness, and coverage, with standard deviations. The Evaluation section now also reports the paired-test support for MAML-Select resource reductions against FedAvg.

Comment 6: *Evaluate fairness across heterogeneous clients using participation fairness, client coverage, or tier-level selection rates.*

Reply 6: We added fairness and coverage metrics. MAML-Select reaches 100% client coverage on all three datasets, so every client participates at least once during the evaluated runs. Its Jain index is between 0.61 and 0.78, below FedAvg and FedGCS. We therefore do not claim that MAML-Select is the fairest selector. The revised claim is narrower: MAML-Select preserves full coverage while improving the compute–energy profile.

Changes incorporated in the revised manuscript: Table II reports Jain fairness and coverage. The Evaluation text discusses participation coverage and fairness behaviour directly in the main manuscript.

Comment 7: *The claims regarding energy efficiency and Green AI are indirect. Moderate the wording or provide stronger evidence.*

Reply 7: We clarified the measurement model while preserving the efficiency evidence. The revised paper reports TFLOPs and model-based energy/carbon estimates derived from stated client-tier assumptions: Tier 1 uses 4 W power and 18 Wh battery capacity, Tier 2 uses 7 W and 30 Wh, and Tier 3 uses 12 W and 48 Wh. Carbon is computed with 475 gCO₂e/kWh. These common assumptions are applied to all methods, so the comparison isolates the effect of client selection.

Changes incorporated in the revised manuscript: The Evaluation section now states the device-tier power, battery, and grid-intensity values used for resource accounting. CodeCarbon v2.4.1 is cited for carbon-accounting metadata and emission units.

Comment 8: *Justify the six state features and provide an ablation study.*

Reply 8: We added a feature rationale and an ablation study. Loss and gradient norm represent statistical utility. Latency and battery represent system constraints. Selection frequency and staleness expose participation imbalance. Removing any one feature changes final Fashion-MNIST accuracy by less than about

0.35 percentage points, indicating that the policy is not fragile to a single missing input. The largest drops occur when battery or loss feedback is removed, showing that both system status and learning feedback are useful. The revised text also explains that frequency and staleness are retained because they support stable participation, even when their removal gives a small accuracy increase in some runs.

Changes incorporated in the revised manuscript: The State Space paragraph explains the role of each feature. The Sensitivity and Ablation section now frames λ sensitivity and feature ablation as complementary tests of the resource-aware objective and the six-feature policy state.

Comment 9: *Fig. 2 has small font sizes and unclear axis scaling.*

Reply 9: We regenerated the figures with larger fonts, boxed axes, clearer legends, and improved placement. The main manuscript now uses compact evidence figures for convergence, efficiency–accuracy trade-off, λ sensitivity, feature ablation, and selector-overhead scaling. The revised Fig. 2 reports CIFAR-100 accuracy, precision, modelled carbon, and test loss over rounds, so the comparison is no longer limited to a single efficiency curve.

Changes incorporated in the revised manuscript: The earlier figure package is replaced by compact main evidence figures plus separate λ -sensitivity and feature-ablation figures, with publication-quality source graphics retained in the package.

Comment 10: *The definition of “Time (Units)” is unclear.*

Reply 10: We removed the ambiguous phrase from the main evidence table. The revised table now reports compute, modelled energy, modelled carbon, Jain fairness, and coverage.

Changes incorporated in the revised manuscript: The Evaluation defines TFLOPs and modelled energy/carbon directly. The ambiguous time-unit row has been removed from the main table.

Comment 11: *Include recent baselines such as CriticalFL and FedGCS.*

Reply 11: We added both methods. FedGCS and CriticalFL are now part of the revised benchmark table, related-work discussion, and main evidence figures. This gives the evaluation eight baselines in total: FedAvg, FedCS, Oort, TiFL, FedCor, FedGCS, CriticalFL, and MAML-Select as the proposed method.

Changes incorporated in the revised manuscript: Related Work, Table I, Table II, and the main evidence figures now include FedGCS and CriticalFL.

Comment 12: *Proofread the manuscript for grammar, punctuation, and sentence structure.*

Reply 12: We completed a full prose pass and shortened the narrative.

Changes incorporated in the revised manuscript: The revised manuscript uses shorter sentences and removes unsupported claims.

Response to Comments of Reviewer 3

Comment 1: *The motivation is not clear.*

Reply 1: We thank the Reviewer for this comment. We sharpened the motivation. The central issue is that client utility is non-stationary: a client useful in one round may be less useful later because the global model, local loss, latency, and participation history change. MAML-Select addresses this by adapting the selection policy from recent feedback before the next decision.

Changes incorporated in the revised manuscript: The Abstract and Introduction now state the non-stationary client-utility problem directly.

Comment 2: *Highlight the novelty and main contribution.*

Reply 2: We now state the contribution more narrowly and clearly. MAML-Select is not a personalization method and not a long-horizon RL selector. It is a first-order meta-learning method for online adaptation of a client-ranking policy.

Changes incorporated in the revised manuscript: The Introduction, Related Work, and Table I now highlight this distinction.

Comment 3: *Add appropriate constraints to the problem formulation.*

Reply 3: We added the available-client set, binary decision variables, cohort-size constraint, and availability constraint.

Changes incorporated in the revised manuscript: The Problem Formulation now enforces $a_{i,t} \in \{0, 1\}$, $\sum_i a_{i,t} = K$, and $a_{i,t} = 0$ for unavailable clients.

Comment 4: *Why is only one method considered as a baseline? Consider at least two more baselines.*

Reply 4: The revised evaluation includes seven baselines: FedAvg, FedCS, Oort, TiFL, FedCor, FedGCS, and CriticalFL.

Changes incorporated in the revised manuscript: The Evaluation and Table II now compare MAML-Select with the expanded baseline set.

Comment 5: *Provide more simulation results to ensure effectiveness.*

Reply 5: We added three datasets, eight compared methods, repeated seeds, standard deviations, paired tests, fairness/coverage metrics, λ sensitivity, feature ablations, and selector-overhead scaling. Fashion-MNIST and CIFAR-10 are reported for 200 rounds, and CIFAR-100 is reported for 150 rounds. The revised evidence therefore covers accuracy, precision, recall, F1, compute, modelled energy/carbon, fairness, coverage, sensitivity, ablation, and server-side selector overhead.

Changes incorporated in the revised manuscript: The main paper includes the compact convergence/trade-off evidence, separate sensitivity and ablation figures, the selector-overhead scaling plot, fairness/coverage

metrics, and standard deviations.

Comment 6: *Remove references from the conclusion and write the conclusion as a concise single paragraph.*

Reply 6: We followed this suggestion.

Changes incorporated in the revised manuscript: The Conclusion is now a single concise paragraph with no citations and one future-work sentence.

Response to Comments of Reviewer 4

Comment 1: *Correct the Impact Statement wording for heterogeneous environments and industrial Internet of Things.*

Reply 1: We adopted the suggested wording and also moderated the impact claims.

Changes incorporated in the revised manuscript: The Impact Statement now uses “heterogeneous environments” and “industrial Internet of Things,” and explains that energy/carbon estimates follow the stated device-tier and grid-intensity assumptions.

Comment 2: *Improve long sentences, grammar, and proofreading throughout the manuscript.*

Reply 2: We revised the manuscript for shorter, clearer prose.

Changes incorporated in the revised manuscript: The Abstract, Impact Statement, Introduction, Related Work, Methodology, Evaluation, and Conclusion were edited for clarity.

Comment 3: *Update the literature review with recent and related papers.*

Reply 3: We updated the Related Work and focused the bibliography. The revised paper retains the original algorithm references for all compared methods and adds recent work relevant to robust selection, adaptive straggler handling, inclusive participation, asynchronous heterogeneity, and non-IID FL.

Changes incorporated in the revised manuscript: Related Work now includes recent IEEE TAI-relevant literature and a focused bibliography.

Comment 4: *Include a comparison table with previous state-of-the-art works.*

Reply 4: We added the requested comparison table.

Changes incorporated in the revised manuscript: Table I compares representative client-selection and meta-learning approaches against MAML-Select.

Comment 5: *Make the Conclusion and Future Work section accurate and concise. Avoid too many citations.*

Reply 5: We agree and revised it accordingly.

Changes incorporated in the revised manuscript: The Conclusion is one paragraph, has no citations, and ends with one concise future-work sentence.

Comment 6: *Fig. 2 requires clarification, justification, and a clearer comparison with other schemes.*

Reply 6: The original figure has been replaced and the presentation has been clarified. The main manuscript now presents convergence, trade-off, λ -sensitivity, feature-ablation, and selector-overhead scaling evidence with clearer axes and captions. The new convergence figure reports multiple round-wise metrics, and the trade-off figure directly positions MAML-Select against the baselines in terms of accuracy change and normalized compute.

Changes incorporated in the revised manuscript: The main evidence, λ -sensitivity, feature-ablation, and selector-scaling figures and captions were regenerated and rewritten.

Measurement Scope

We keep the energy/carbon scope explicit without weakening the efficiency contribution. The reported values are model-based estimates tied to the stated client-tier power and grid-intensity assumptions. The same accounting model is applied to every method, making the resource comparison reproducible and fair.